Anirudh Jakhmola

PhD Researcher | Phage Biology & Cryo-EM

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Education

University of Sheffield, Microbiology / Structural Biology — Sheffield, UK
- MRC DiMeN Doctoral Training Partnership scholarship recipient — Oct 2023 – Oct 2027
- Supervised by Dr. Robert Fagan (Fagan Lab)

HBNI: National Institute of Science Education and Research, Biology (Minor: Physics) — Bhubaneswar, India, May 2018 – May 2023
- DISHA Scholarship, Department of Atomic Energy

DAV Public School, Science — Kotdwara, India
- DAV Merit Recognition Award — Aug 2016 – May 2017

Experience

Fagan Lab, University of Sheffield, PhD Researcher — Sheffield, UK
- Engineering C. difficile bacteriophages to combat gut infections — Oct 2023 – present
- Using TEM, Cryo-EM, and structure-guided tools to understand phage structure and host interactions — 3 years 1 month

University of Sheffield, Graduate Teaching Assistant (Part-time) — Sheffield, UK
- Taught 6 undergraduate courses across two academic years — Dec 2023 – Sept 2025, 1 year 10 months

SciAstra Education, Program Manager (Part-time, Remote) — NOIDA, India
- Lead workshops and classes for various scientific courses — June 2023 – present
- Managed 10+ team members across US, UK, EU, and India — 3 years 6 months

Cellular Cytoskeletal Dynamics Lab, NISER, Masters Dissertation Student — Bhubaneswar, India
- Characterisation of bacterial ParM proteins in a heterologous system — June 2021 – May 2023, 2 years
- Employed microscopy and FRAP techniques using DeltaVision and Leica microscopes for analysis of plasmid segregation proteins
- Generated ATP-sensitive mutants of MreB protein and cloned them into a heterologous system

Max Planck Institute for Evolutionary Biology, Research Intern — Plön, Germany
- Standardised the procedure for generating site-specific mutants of ΦX174 virus by modifying the current protocol used for E. coli plasmids — May 2022 – Aug 2022, 4 months
- Tested evolved ΦX174 viral lysates against E. coli C, K-12, E. coli B and other strains for cross-infection screening
- Troubleshot the genotype of evolved ΦX174 phages

National Institute of Science Education and Research, Research Intern — Bhubaneswar, India
- Tested Drosophila trachea for apoptosis during larval stage transitions — May 2019 – Aug 2019, 4 months
- Characterised eight proteins in Drosophila using western blotting and antibody staining
- Reading project on six techniques for determining respiratory substrate usage by an organism

Volunteer

Florey Institute of Infection, University of Sheffield, Management Team Member — Sheffield, UK, Sept 2024 – present

PGR Core Committee, University of Sheffield, Inclusions and Diversity Officer — Sheffield, UK, Sept 2024 – Apr 2025

NISER, Bhubaneswar, Organising Committee, CEMBIOS 2023 — Bhubaneswar, India, Feb 2023 – Feb 2023

NISER, Bhubaneswar, Science Demonstrator, NISER Open Day — Bhubaneswar, India, Mar 2019 – Mar 2019

NISER, Bhubaneswar, Audio Manager, Drama and Music Club — Bhubaneswar, India, Sept 2018 – Dec 2022

NISER, Bhubaneswar, Podcast Manager, Science Activities Club — Bhubaneswar, India, Dec 2018 – Dec 2022

NISER, Bhubaneswar, Social Media Outreach Coordinator, Art Club — Bhubaneswar, India, July 2019 – Dec 2021

Awards

SURE Project Grant — Apr 2025
Secured funding to hire a student to work on C. difficile phage infection research
University of Sheffield

Best Talk Award, UK Cellular Microbiology — July 2024
University of Sheffield

iGEM Synthetic Biology Gold Medal — Oct 2024
Advisor for Phosphix team on engineering an enzyme for sustainable fertiliser production
iGEM Foundation

MRC DiMeN DTP Scholarship — Oct 2023
Funding awarded for at least four years for a PhD in the DiMeN Doctoral Training Partnership
Medical Research Council

DISHA Scholarship — Aug 2018
Monthly scholarship and contingency grant to top 1% STEM students; held for five years (2018–2023)
Department of Atomic Energy, India

DAV Merit Recognition — Dec 2018
Merit-based award to encourage excellence at DAV Public School
DAV Public School

Educational Award — June 2017
Merit-based award for excellent achievers in their senior year
BEL, Ministry of Defence, India

Conferences and Presentations

- **Microbiology Society Annual Conference 2026** (talk and poster), Belfast, UK, Apr 2026
- **Viruses of Microbes UK 2026** (talk and poster), Newcastle, UK, Jan 2026
- **ClostPath 2025**, 14th International Conference on the Molecular Biology and Pathogenesis of Clostridia (poster and session chair), Institut Pasteur, Paris, France, Sep 2025
- **Cross-Channel C. difficile Forum 2025** (talk), Newcastle, UK, Jun 2025
- **UK Cellular Microbiology** (Best Talk Award), University of Sheffield, UK, Jul 2024
- **Viruses of Microbes UK**, Northumbria University, UK, Jan 2024
- **NEPG Symposium 2023**, Newcastle University, UK, Nov 2023
- **CEMBIOS 2023** (Cryo-Electron Microscopy), NISER, Bhubaneswar, India, Feb 2023
- **Genome Structural and Functional Annotation Workshop**, University of Connecticut, USA, Nov 2022
- **Biostatistics: A User's Perspective Workshop**, IISER Pune, India, Feb 2022
- **Cardiovascular Diseases: New Advances and Translational Applications Workshop**, National University of Singapore, Sep 2021
- **International Conference for Autophagy and Lysosomes**, Indian Institute of Science, Bangalore, Feb 2020

Publications

A set of genetic tools for use in Clostridioides difficile and related species Feb 2026
Microbiology 172(2):001665 (2026). † Joint first authors.
H. Fisher†, L. de Lussy Kubisa†, ***Anirudh Jakhmola†***, E. Walker, J. A. Kirk, P. Oatley, R. R. Chaudhuri, G. R. Douce, M. J. Ormsby, R. P. Fagan
10.1099/mic.0.001665 (Microbiology)

Dynamic filament formation by bacterial actin homologs (MS thesis) May 2023
Integrated BS-MS dissertation, supervised by Dr. Ramanujam Srinivasan
Anirudh Jakhmola
(MS thesis, NISER Bhubaneswar)

A role for the last C-terminal helix of the F plasmid segregating protein SopA in nucleoid binding and plasmid maintenance Jan 2022
Plasmid 119-120:102617 (2022)
D. Mishra, ***Anirudh Jakhmola***, R. Srinivasan
10.1016/j.plasmid.2022.102617 (Plasmid)

Skills

Lab Skills

Programming

Tools

Languages

English
Fluent (TOEFL-iBT Score 107)

Hindi
Native speaker

Garhwali
Native speaker

Odia
Elementary

German
Elementary

References

Available upon request